



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to MADURA SUNDARARAJ FUEL STATION

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

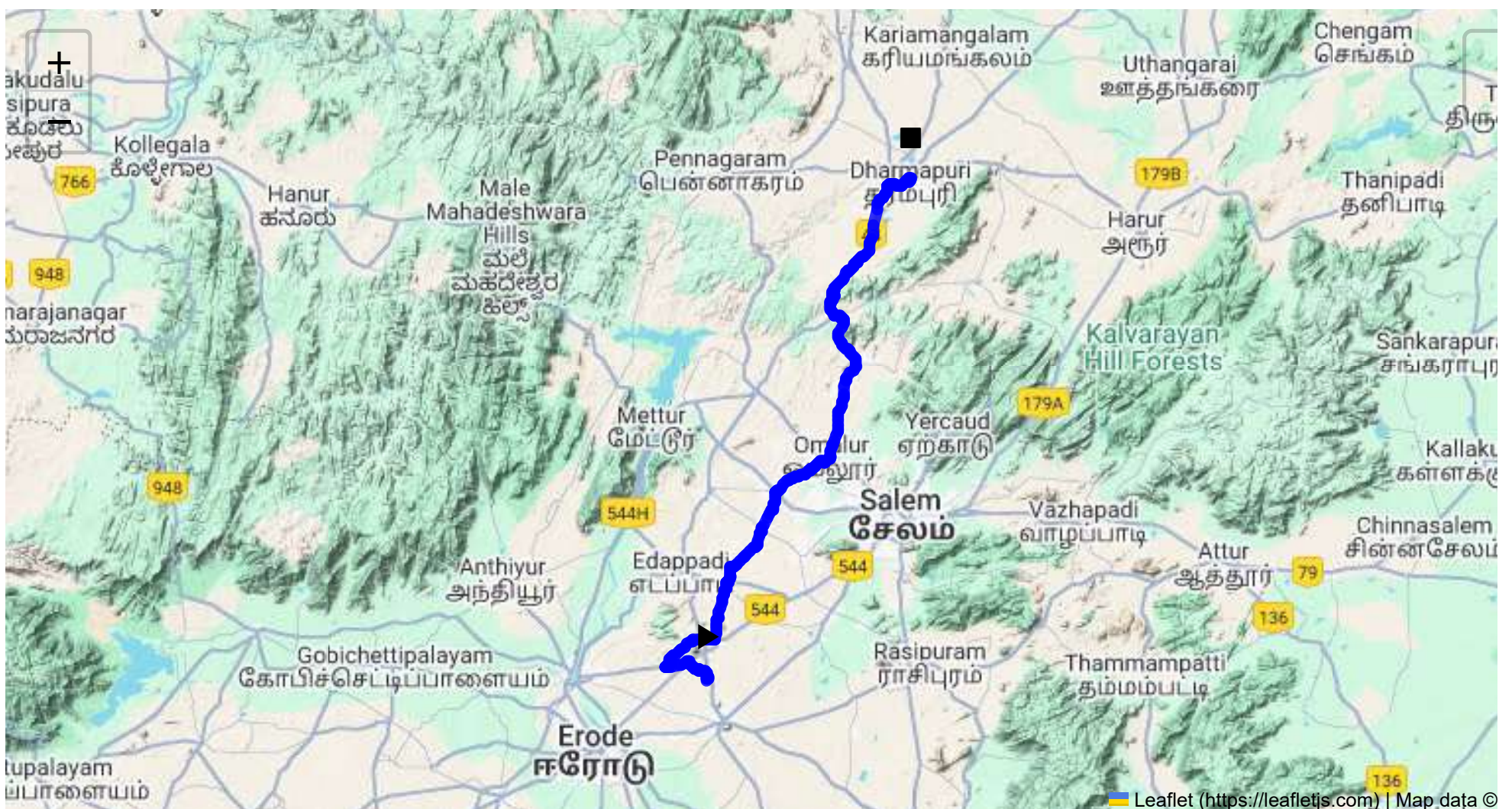
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 106.54 km
Estimated Duration: 2.4 hours
Adjusted Duration (Heavy Vehicle): 3.0 hours
Start: (11.4381, 77.8734)
End: (12.118672, 78.156859)



Welcome to the Journey Risk Management Study

Certainly! Here is a detailed analysis of the route from Sangagiri to Dharmapuri in Tamil Nadu, India, focusing on the safety and risk considerations for truck drivers carrying hazardous materials.

1. Overview of the Route Map:

The route is approximately 106.54 kilometers in length, traversing from Sangagiri through various towns, including Sankari, Padaiveedu, and Tharamangalam before reaching Dharmapuri. The route travels primarily via SH 86 and involves both rural and semi-urban stretches.

2. Typical Weather Conditions and Hazards:

- **Weather Patterns:** Tamil Nadu experiences a tropical climate with hot summers, monsoons, and mild winters. The region is typically hottest from March to June.
- **Monsoon Warnings:** Heavy rains during the monsoon season (June to September) may result in water-logged roads, reducing visibility and traction.

3. Traffic Patterns:

- **Peak Hours:** Generally, peak traffic hours are from 8:00 AM to 10:00 AM and from 5:00 PM to 7:00 PM due to local commuting.
- **Congestion Areas:** Potential congestion could occur near major junctions and towns, especially Sankari and Tharamangalam. Local markets may also cause delays.

4. Road Quality and Infrastructure:

- **Road Quality:** The SH 86 stretches are generally maintained but may exhibit wear, potholes, or repairs particularly during and after monsoons.
- **Infrastructure:** Be aware of narrow bridges and minor elevation changes which could impact larger vehicles.

5. Suggestions for Alternative Routes:

- In emergencies, consider taking NH 544 and NH 44 which are wider and more direct but could be longer in terms of distance.

6. Local Regulations on Hazardous Material Transport:

- **Regulations:** Follow local norms for carrying hazardous materials, ensuring all goods are clearly labeled, and that transport permits are in place. Restrictions may apply especially near populated or environmentally sensitive areas.

7. Historical Incidents:

- **Accidents:** Data on past accidents, particularly involving heavy trucks, should be reviewed. Areas previously affected could include congested town roads and during the monsoon when visibility is reduced.

8. Environmental Considerations:

- **Sensitive Areas:** Sensitive regions such as agricultural zones should be navigated with care to prevent environmental contamination in case of spills.

9. Communication Coverage:

- **Potential Dead Zones:** Rural stretches between towns may have limited mobile reception. Ensure communication devices are working, and maps are downloaded for offline use.

10. Estimated Emergency Response Times:

- **Urban Areas:** Approximately 15-30 minutes depending on proximity to services in towns like Tharamangalam.
- **Rural Areas:** Could extend up to 45 minutes or more due to lesser accessibility.

12. Overall Summary of Risk Assessment:

- **Low to Moderate Risk:** For most of the route with specific precautions.
- **High Risk:** During monsoon conditions, congested areas, or winding segments of SH 86.
- **Recommendations:** Prepare for potential road repairs, follow up-to-date traffic advisories, ensure all safety equipment and materials for hazardous goods are compliant, and plan for contingencies.

This assessment highlights the importance of staying informed and prepared for varying conditions that could impact transportation safety on this route. Regular updates from regional travel advisories and continuous communication with local transport authorities should also be considered essential.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	Medium	11.44900, 77.87406	30 KM/Hr
3	Turn	High	11.45352, 77.85698	15 KM/Hr
4	Turn	Medium	11.45847, 77.85140	30 KM/Hr
5	Turn	Medium	11.45898, 77.85135	30 KM/Hr
6	Turn	High	11.46303, 77.84945	15 KM/Hr
7	Turn	Medium	11.45571, 77.81702	30 KM/Hr
8	Turn	High	11.45615, 77.81677	15 KM/Hr
9	Blind Spot	Blind Spot	11.49395, 77.88537	10 KM/Hr
10	Turn	High	11.49465, 77.88488	15 KM/Hr
11	Turn	High	11.56096, 77.89931	15 KM/Hr
12	Turn	High	11.57035, 77.90223	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
13	Turn	High	11.57041, 77.90221	15 KM/Hr
14	Turn	Medium	11.58279, 77.90658	30 KM/Hr
15	Turn	Medium	11.63833, 77.94818	30 KM/Hr
16	Turn	Medium	11.64171, 77.95184	30 KM/Hr
17	Turn	Medium	11.64243, 77.95198	30 KM/Hr
18	Turn	Medium	11.64721, 77.95118	30 KM/Hr
19	Turn	High	11.69595, 77.97114	15 KM/Hr
20	Turn	Medium	11.71386, 77.99552	30 KM/Hr
21	Turn	Medium	11.71735, 78.00019	30 KM/Hr
22	Turn	Medium	11.72765, 78.02291	30 KM/Hr
23	Turn	Medium	11.73786, 78.03505	30 KM/Hr
24	Turn	High	11.73652, 78.03866	15 KM/Hr
25	Turn	Medium	11.74110, 78.04575	30 KM/Hr
26	Turn	Medium	11.74257, 78.04708	30 KM/Hr
27	Turn	Medium	11.87022, 78.08380	30 KM/Hr
28	Turn	Medium	11.87913, 78.08206	30 KM/Hr
29	Turn	Medium	11.92566, 78.06503	30 KM/Hr
30	Turn	Medium	11.95810, 78.04971	30 KM/Hr
31	Turn	Medium	11.95865, 78.04991	30 KM/Hr
32	Turn	Medium	11.95963, 78.04954	30 KM/Hr
33	Turn	Medium	11.96072, 78.04963	30 KM/Hr
34	Turn	Medium	11.96281, 78.05124	30 KM/Hr
35	Turn	Medium	11.96384, 78.05141	30 KM/Hr
36	Turn	Medium	11.96442, 78.05104	30 KM/Hr
37	Turn	High	11.96557, 78.04930	15 KM/Hr
38	Turn	High	11.96657, 78.04930	15 KM/Hr
39	Turn	Medium	12.10459, 78.12492	30 KM/Hr
40	Turn	High	12.11584, 78.12772	15 KM/Hr
41	Turn	Medium	12.11561, 78.12856	30 KM/Hr
42	Turn	Medium	12.11466, 78.14396	30 KM/Hr
43	Turn	Medium	12.11485, 78.14484	30 KM/Hr
44	Turn	High	12.11431, 78.14671	15 KM/Hr
45	Turn	Medium	12.12199, 78.15528	30 KM/Hr
46	Turn	High	12.12519, 78.15694	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
47	Turn	High	12.12478, 78.15851	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	clinic	Sun Eye Care	11.6936242, 77.9730948	30 km/h	Medium
3	hospital	Dr. Pandiyarajan Hospital	11.692704, 77.969344	30 km/h	Medium
4	hospital	Dr. Sengodan Hospital	11.6959527, 77.9695325	30 km/h	Medium
5	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
6	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
7	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
8	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
9	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
10	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
11	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
12	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
14	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
15	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
16	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
17	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
18	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
19	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
20	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
21	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
22	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
23	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
24	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
25	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
26	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
28	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
30	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
31	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
32	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
34	clinic	Ayisha Ayurvedic Clinic	12.1203099, 78.1514403	30 km/h	Medium
35	hospital	Thangam, Hospital	12.1191829, 78.1566752	30 km/h	Medium
36	hospital	Pranitha Hospital	12.12133, 78.154418	30 km/h	Medium
37	hospital	Ammakannu Hospital	12.119828, 78.1566101	30 km/h	Medium
38	hospital	Dharmapuri Medl College Hospital	12.12341, 78.1576161	30 km/h	Medium
39	hospital	Dnv Ortho Care	12.12005, 78.157093	30 km/h	Medium
40	hospital	Dr. Agarwals Eye Hospital , Dharmapuri	12.1252773, 78.1582779	30 km/h	Medium
41	clinic	Janani Maternity and Child Health Clinic	12.1251513, 78.1582696	30 km/h	Medium
42	hospital	Dharmapuri Cancer Centre	12.1252142, 78.1574084	30 km/h	Medium
43	clinic	M.K Speciality Clinic	12.1252999, 78.1583306	30 km/h	Medium
44	hospital	Kala Eye Hospital	12.1208845, 78.1582225	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
45	hospital	Nithya Hospital	12.1244251, 78.1587231	30 km/h	Medium
46	hospital	K.V. Hospital - Senthil Kidney Centre	12.1255592, 78.1581934	30 km/h	Medium
47	clinic	Vinayaga Poly Clinic	12.1255515, 78.15816	30 km/h	Medium
48	hospital	Usha Hospital	12.1259805, 78.1586924	30 km/h	Medium
49	clinic	Alex Ortho Care	12.125915, 78.1586122	30 km/h	Medium
50	hospital	Guru Hospital	12.1229939, 78.1587116	30 km/h	Medium
51	hospital	Sri Ramakrishna Hospital	12.1271589, 78.1582873	30 km/h	Medium
52	clinic	Ramesh Clinic	12.1272995, 78.1582422	30 km/h	Medium
53	hospital	Ssk Hospital	12.1188726, 78.1565367	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
13	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
27	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
29	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
33	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45847, 77.85140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45898, 77.85135



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45571, 77.81702



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45615, 77.81677



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49395, 77.88537



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49465, 77.88488



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.56096, 77.89931



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.57035, 77.90223



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.57041, 77.90221



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.58279, 77.90658



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63833, 77.94818



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64171, 77.95184



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64243, 77.95198



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64721, 77.95118



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.69595, 77.97114



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71386, 77.99552



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71735, 78.00019



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.72765, 78.02291



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73786, 78.03505



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73652, 78.03866



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74110, 78.04575



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74257, 78.04708



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.87022, 78.08380



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.87913, 78.08206



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.92566, 78.06503



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.95810, 78.04971



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.95865, 78.04991



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

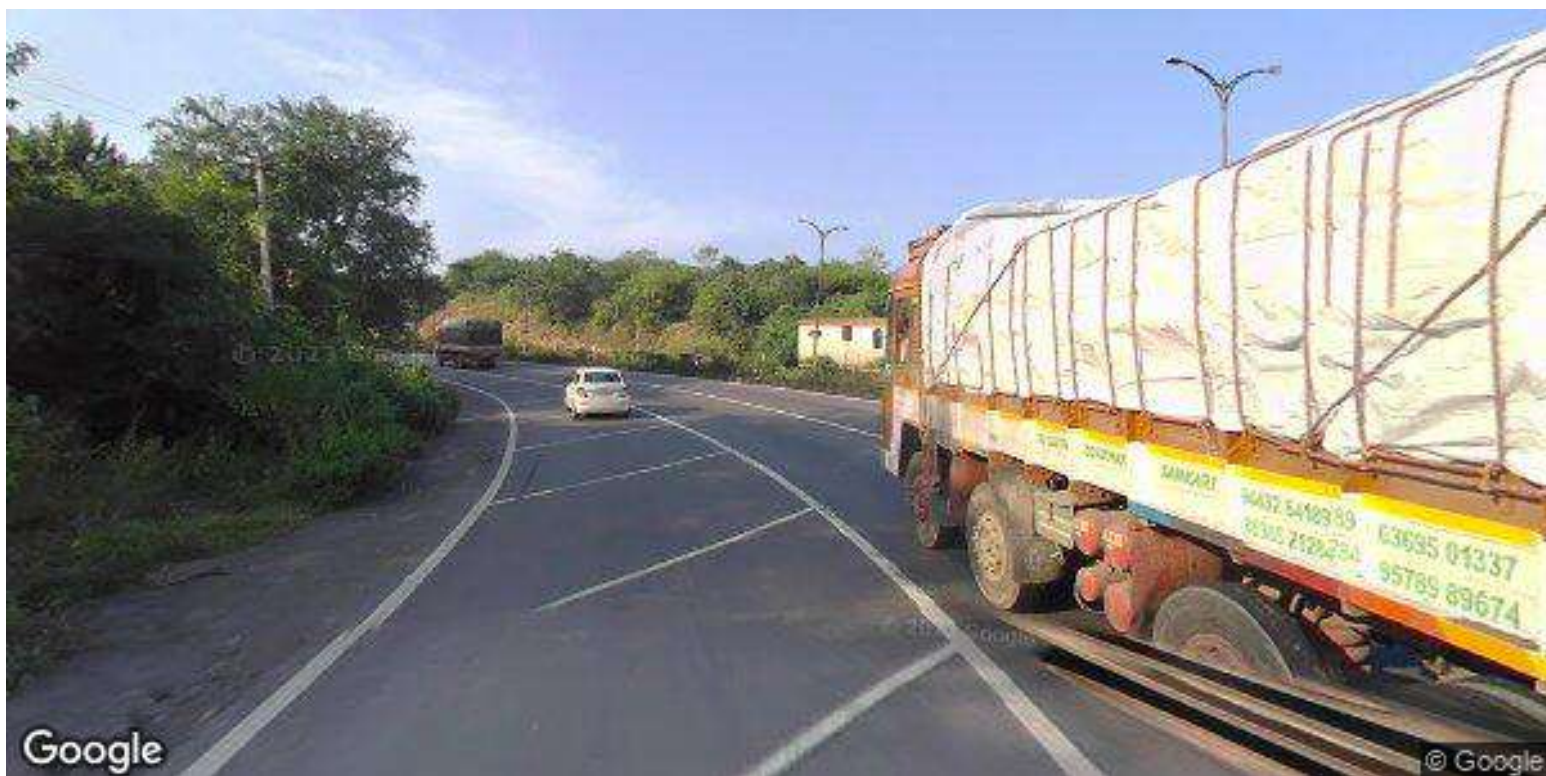
Coordinates: 11.95963, 78.04954



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96072, 78.04963



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96281, 78.05124



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96384, 78.05141



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96442, 78.05104



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96557, 78.04930



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96657, 78.04930



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.10459, 78.12492



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.11584, 78.12772



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.11561, 78.12856



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.11466, 78.14396



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.11485, 78.14484



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.11431, 78.14671



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.12199, 78.15528



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.12519, 78.15694



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.12478, 78.15851
